TEST REPORT

Product	:	Seeed Studio XIAO ESP32S3, Seeed Studio XIAO ESP32S3 Sense
Trade mark	:	Seeed Studio
Model/Type reference	:	XIAO-ESP32-S3, XIAO-ESP32-S3-Sense
Serial Number	:	N/A
Report Number	:	EED32P80440402
Date of Issue	:	Apr. 24, 2023
Test Standards	:	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-17 V3.2.4 (2020-09)
Test result	:	PASS

Prepared for:

Seeed Technology Co., Ltd
9F, Building G3, TCL International E city,
Zhongshanyuan Road,
Nanshan, Shenzhen, China.

Prepared by:

Centre Testing International Group Co., Ltd.
Hongwei Industrial Zone, Bao'an 70 District,
Shenzhen, Guangdong, China
TEL: +86-755-3368 3668
FAX: +86-755-3368 3385

Compiled by: Wifi.Lei Reviewed by: Deng Binbin

Approved by: Aaron Ma Date of issue: Apr. 24, 2023

Aaron Ma Check No.:1452310323

2 Version

Version No.	Date	Description
00	Apr. 24, 2023	Original

3 Test Summary

Electromagnetic Compatibility (EMC) Part				
Electromagnetic Interference (EMI)				
Test	**Test Requirement**	**Test Method**	**Limit**	**Result**
Radiated Emission	ETSI EN 301 489-17 V3.2.4 (2020-09) Clause 7.1	EN 301 489-1 V2.2.3 (2019-11) Clause 8.2	Clause 8.2.3	PASS
Conducted Emission (DC port)	ETSI EN 301 489-17 V3.2.4 (2020-09) Clause 7.1	EN 301 489-1 V2.2.3 (2019-11) Clause 8.3	Clause 8.3.3	N/A[1]
Conducted Emission (AC port)	ETSI EN 301 489-17 V3.2.4 (2020-09) Clause 7.1	EN 301 489-1 V2.2.3 (2019-11) Clause 8.4	Clause 8.4.3	N/A[4]
Harmonic Emission on AC, 50Hz to 2kHz	ETSI EN 301 489-17 V3.2.4 (2020-09) Clause 7.1	EN 301 489-1 V2.2.3 (2019-11) Clause 8.5	Clause 8.5	N/A[2]
Flicker Emission on AC	ETSI EN 301 489-17 V3.2.4 (2020-09) Clause 7.1	EN 301 489-1 V2.2.3 (2019-11) Clause 8.6	Clause 8.6	N/A[4]
Conducted Emission (telecommunication port)	ETSI EN 301 489-17 V3.2.4 (2020-09) Clause 7.1	EN 301 489-1 V2.2.3 (2019-11) Clause 8.7	Clause 8.7.3	N/A
Electromagnetic Susceptibility(EMS)				
ESD (Electrostatic Discharge)	ETSI EN 301 489-17 V3.2.4 (2020-09) Clause 7.2	EN 301 489-1 V2.2.3 (2019-11) Clause 9.3	Clause 9.3.3	PASS
Radiated Immunity, 80MHz to 6 GHz	ETSI EN 301 489-17 V3.2.4 (2020-09) Clause 7.2	EN 301 489-1 V2.2.3 (2019-11) Clause 9.2	Clause 9.2.3	PASS
EFT (Electrical Fast Transients)	ETSI EN 301 489-17 V3.2.4 (2020-09) Clause 7.2	EN 301 489-1 V2.2.3 (2019-11) Clause 9.4	Clause 9.4.3	N/A[4]
Surge Immunity	ETSI EN 301 489-17 V3.2.4 (2020-09) Clause 7.2	EN 301 489-1 V2.2.3 (2019-11) Clause 9.8	Clause 9.8.3	N/A[4]
Injected Currents 150kHz to 80MHz	ETSI EN 301 489-17 V3.2.4 (2020-09) Clause 7.2	EN 301 489-1 V2.2.3 (2019-11) Clause 9.5	Clause 9.5.3	N/A[4]
Voltage Dips and Interruptions	ETSI EN 301 489-17 V3.2.4 (2020-09) Clause 7.2	EN 301 489-1 V2.2.3 (2019-11) Clause 9.7	Clause 9.7.3	N/A[4]
Transients and Surges in the Vehicular Environment	ETSI EN 301 489-17 V3.2.4 (2020-09) Clause 7.2	EN 301 489-1 V2.2.3 (2019-11) Clause 9.6	Clause 9.6.3	N/A[3]

Remark:

Company Name and Address shown on Report, the sample(s) and sample Information was/ were provided by the applicant who should be responsible for the authenticity which CTI hasn't verified.

Tx: In this whole report Tx (or tx) means Transmitter.
Rx: In this whole report Rx (or rx) means Receiver.
RF: In this whole report RF means Radiated Frequency.
CH: In this whole report CH means channel.
Volt: In this whole report Volt means Voltage.
Temp: In this whole report Temp means Temperature.
Humid: In this whole report Humid means humidity.
Press: In this whole report Press means Pressure.
N/A: In this whole report not application.
N/A[1)] The tested sample has no DC mains input/output port , therefore it is not applicable.
N/A[2)] The Product belongs to Class A, and its power is less than 75W, so it deems to fulfil this standard without testing.
N/A[3)] The tested sample is not used in the vehicle, therefore it is not applicable.
N/A[4)] The tested sample has Battery, therefore it is not applicable.

Model No.: XIAO-ESP32-S3, XIAO-ESP32-S3-Sense

XIAO-ESP32-S3-Sense is composed of XIAO-ESP32-S3 main board and expansion board. XIAO-ESP32-S3 is a single main board without an expansion board, and compared with the above-mentioned XIAO-ESP32-S3 Sense main board, there is no connector on the board. Their electrical circuit design, layout, components used and internal wiring are identical.

4 Contents

Page

1 COVER PAGE 1

2 VERSION 2

3 TEST SUMMARY 3

4 CONTENTS 5

5 GENERAL INFORMATION 6

5.1 CLIENT INFORMATION 6
5.2 GENERAL DESCRIPTION OF EUT 6
5.3 PRODUCT SPECIFICATION SUBJECTIVE TO THIS STANDARD 6
5.4 OTHER INFORMATION 6
5.5 DESCRIPTION OF SUPPORT UNITS 6
5.6 TEST LOCATION 7
5.7 DEVIATION FROM STANDARDS 7
5.8 ABNORMALITIES FROM STANDARD CONDITIONS 7
5.9 OTHER INFORMATION REQUESTED BY THE CUSTOMER 7
5.10 MONITORING OF EUT FOR THE IMMUNITY TEST 7
5.11 MEASUREMENT UNCERTAINTY (95% CONFIDENCE LEVELS, K=2) 7

6 EQUIPMENT LIST 8

7 EMC REQUIREMENTS SPECIFICATION IN EN 301 489-17 10

7.1 EMI (EMISSION) 10
7.1.1 Radiated Emission *10*
7.2 EMS (IMMUNITY) 15
7.2.1 Radiated Immunity *16*
7.2.2 ESD *18*

APPENDIX PHOTOGRAPHS OF TEST SETUP 20

APPENDIX PHOTOGRAPHS OF EUT 23

5 General Information

5.1 Client Information

Applicant:	Seeed Technology Co., Ltd
Address of Applicant:	9F, Building G3, TCL International E city, Zhongshanyuan Road, Nanshan, Shenzhen, China.
Manufacturer:	Seeed Technology Co., Ltd
Address of Manufacturer:	9F, Building G3, TCL International E city, Zhongshanyuan Road, Nanshan, Shenzhen, China.
Factory :	Shenzhen Xinxian Technology Co., Limited
Address of Factory:	F5, Building B17, Hengfeng Industrial City, No. 739 Zhoushi Rd, Baoan District, Shenzhen, Guangdong, P.R.C.

5.2 General Description of EUT

Product Name:	Seeed Studio XIAO ESP32S3, Seeed Studio XIAO ESP32S3 Sense
Model No.(EUT):	XIAO-ESP32-S3, XIAO-ESP32-S3-Sense
Test Model:	XIAO-ESP32-S3-Sense
Trade Mark:	Seeed Studio

5.3 Product Specification subjective to this standard

Power Supply:	DC 5.0V
Test Voltage:	DC 5.0V
Test Mode:	
BT mode &WIFI mode	The EUT wireless linked to the Notebook, exchanging the data.
Standby mode:	Keep the EUT in standby mode.

5.4 Other Information

RED Directive:	2014/53/EU
Sample Received Date:	Mar. 31, 2023
Sample tested Date:	Mar. 31, 2023 to Apr. 14, 2023

5.5 Description of Support Units

The EUT has been tested with associated equipment below.

Description	Manufacturer	Model No.	Certification	Supplied by
---	---	---	---	---

5.6 Test Location

All tests were performed at:

Centre Testing International Group Co., Ltd.
Hongwei Industrial Zone, Bao'an 70 District, Shenzhen, Guangdong, China
Telephone: +86 (0) 755 3368 3668 Fax:+86 (0) 755 3368 3385
No tests were sub-contracted.

5.7 Deviation from Standards

None.

5.8 Abnormalities from Standard Conditions

None.

5.9 Other Information Requested by the Customer

None.

5.10 Monitoring of EUT for the Immunity Test

Visual: Monitoring the notebook screen and the status of EUT.

Audio: N/A.

5.11 Measurement Uncertainty (95% confidence levels, k=2)

No.	Item	Measurement Uncertainty
1	Conduction emission	3.5dB (9kHz to 150kHz)
		3.1dB (150kHz to 30MHz)
2	Radiated emission	4.9dB (30MHz-1GHz)
		4.7dB (1GHz-6GHz)
3	Temperature test	0.64°C
4	Humidity test	3.8%
5	DC power test	0.026%

6 Equipment List

3M Semi-anechoic Chamber (2)- Radiated emissions Test					
No.	**Equipment**	**Manufacturer**	**Model**	**Serial No.**	**Due Date**
1	TRILOG Broadband Antenna	Schwarzbeck	VULB9163	9163-484	04/28/2023
2	3M Chamber & Accessory Equipment	TDK	SAC-3	/	05/21/2025
3	Receiver	R&S	ESCI7	100938-003	09/27/2023
4	Microwave Preamplifier	JS Tonscend	EMC051845SE	980380	12/22/2023
5	Horn Antenna	Schwarzbeck	BBHA 9120D	9120D-1869	04/16/2024

Shielding Room No. 1 - Electrostatic discharge (ESD) Test (IEC 61000-4-2)					
No.	**Equipment**	**Manufacturer**	**Model**	**Serial No.**	**Due Date**
1	ESD Simulator	TESEQ	NSG437	1182	06/09/2023

3M Full-anechoic Chamber - Continuous RF electromagnetic radiated field disturbances Test (IEC 61000-4-3)					
No.	**Equipment**	**Manufacturer**	**Model**	**Serial No.**	**Due Date**
1	3M Chamber & Accessory Equipment	ETS-LINDGREN	FACT-3	3510	05/19/2025
2	Stacked double Log.-Per. Antenna	SCHWARZBECK	STLP9128	9128ES-110	03/30/2024
3	Signal Generator	R&S	SMB 100B	103084	05/19/2023
4	Power Probe	R&S	NRP6A	103342	07/11/2023
5	Power Probe	R&S	NRP6A	103343	07/11/2023
6	Power Amplifier	R&S	BBA 150-BC500	104743	06/05/2023
7	Power Amplifier	R&S	BLMA 1060-100	2113427	07/05/2023
8	RF switch	R&S	OSP220	102205	---
9	Horn Antenna	R&S	STLP 9149	0776	05/21/2024
10	Directional coupler	BONN	BDC 1060-40/500	2128343-04	11/27/2023

7 EMC Requirements Specification in EN 301 489-17

EMI in EN 301 489-1, sub clause 7.1 table 1.

EMS in EN 301 489-1, sub clause 7.2 table 2.

7.1 EMI (Emission)

7.1.1 Radiated Emission

Test Requirement: EN 301 489-17 Clause 7.1
Test Method: EN 301 489-1 Clause 8.2.2
EUT Operation:

Ambient: Temp.: 22°C Humid.: 53% Press.: 1010 mbar

Test Mode: BT mode & WiFi mode

Test Status: The test worst case mode is recorded in the report.

Receive Setup:

Frequency range (MHz)	Detector	RBW	VBW
30-1000	Quasi-peak	120kHz	300kHz
Above 1000	Peak	1MHz	3MHz

Limit:

Frequency	Limit(@3m)	Remark
30MHz-230MHz	40dBuv/m	QP value
230MHz-1GHz	47dBuv/m	QP value
1GHz-3GHz	50dBuv/m	Average value
	70dBuv/m	PK value
3GHz-6GHz	54dBuv/m	Average value
	74dBuv/m	PK value

Test Setup:

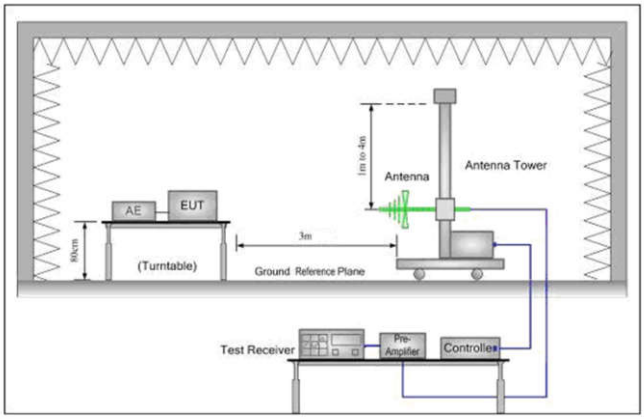


Figure 1. 30MHz to 1GHz

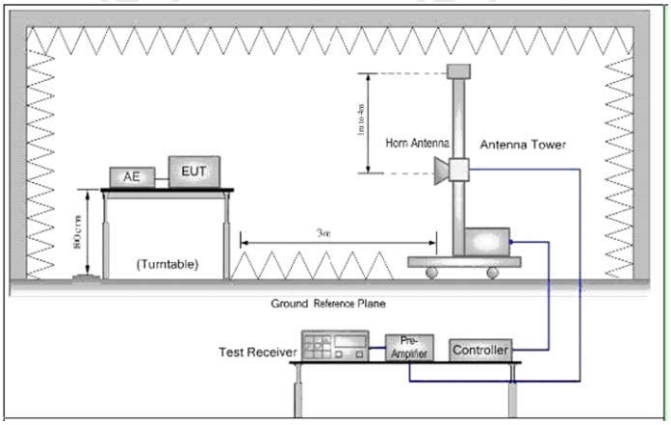


Figure 2. Above 1 GHz

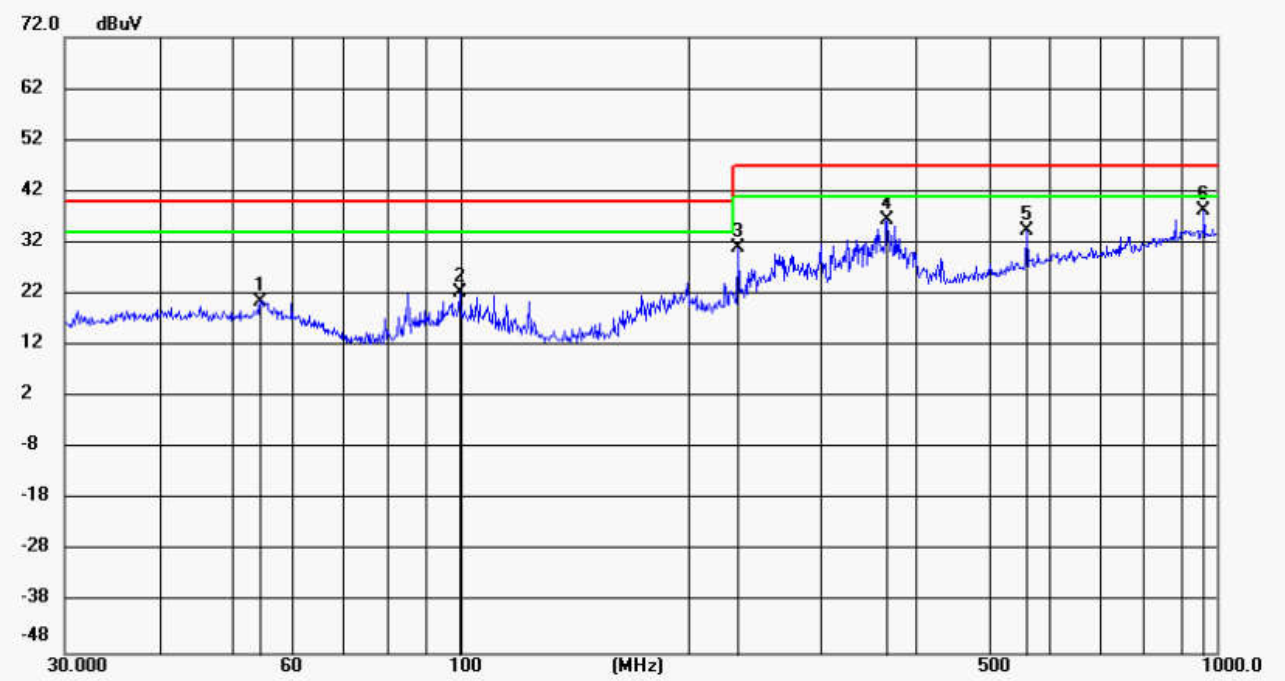
Test Procedure:

1. From 30 MHz to1GHz test procedure as below:
1) The radiated emissions were tested in a semi-anechoic chamber.
2) The EUT is placed on a turntable, which is 0.8m above ground plane.
3) The turntable shall be rotated for 360 degrees to determine the position of maximum emission level.
4) EUT is set 3m away from the receiving antenna, which is moved from 1m to 4m to find out the maximum emissions.
5) Maximum procedure was performed on the six highest emissions to ensure EUT compliance.
6) And also, each emission was to be maximized by changing the polarization of receiving antenna both horizontal and vertical.
7) Repeat above procedures until the measurements for all frequencies are complete.
2. Above 1GHz test procedure as below:
1) Different between above is the test site, change from Semi- Anechoic Chamber to fully Anechoic Chamber

Equipment Used: Refer to section 6 for details.

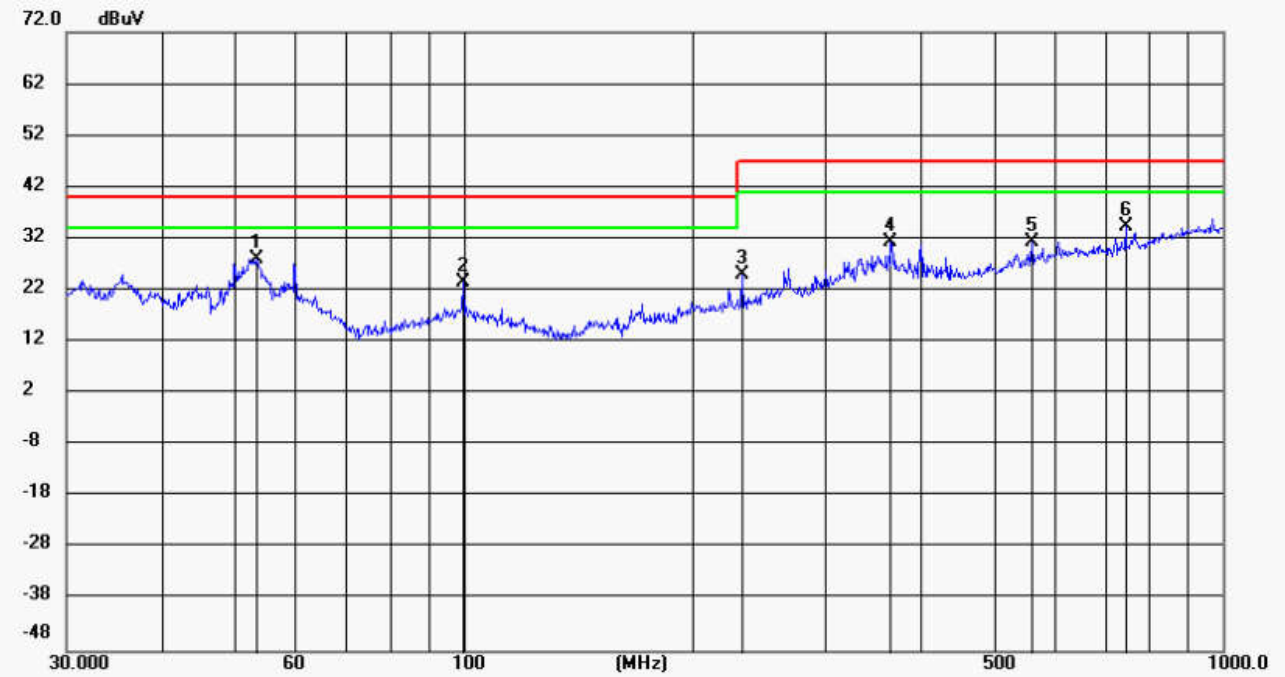
Test result: PASS

Measurement Data（XIAO-ESP32-S3-Sense ）:
Below 1GHz（QP）:
Horizontal



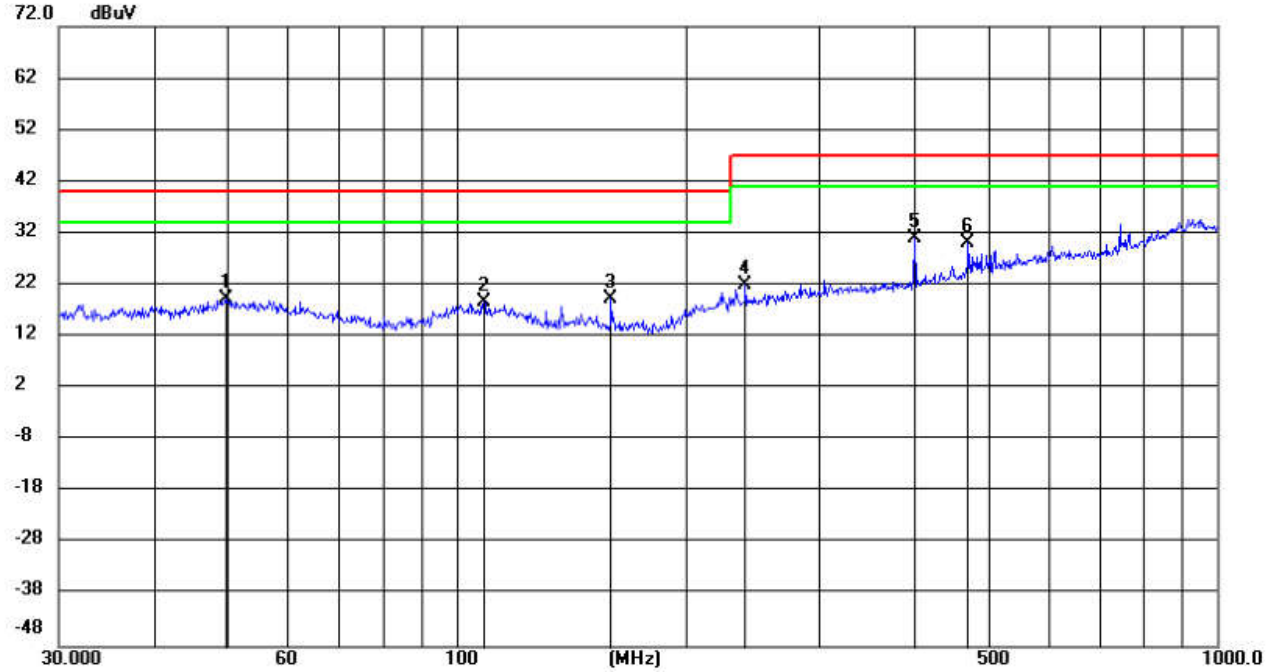
No.	Mk.	Freq.	Reading Level	Correct Factor	Measure-ment	Limit	Margin		Antenna Height	Table Degree	
		MHz	dBuV	dB	dBuV/m	dBuV/m	dB	Detector	cm	degree	Comment
1		54.4515	6.75	13.95	20.70	40.00	-19.30	QP	100	39	
2		99.8777	8.27	14.03	22.30	40.00	-17.70	QP	200	4	
3		233.3486	16.12	14.93	31.05	47.00	-15.95	QP	100	103	
4		366.8231	17.76	18.68	36.44	47.00	-10.56	QP	100	264	
5		560.6928	11.20	23.05	34.25	47.00	-12.75	QP	200	247	
6	*	962.1622	9.60	28.73	38.33	47.00	-8.67	QP	100	232	

Vertical



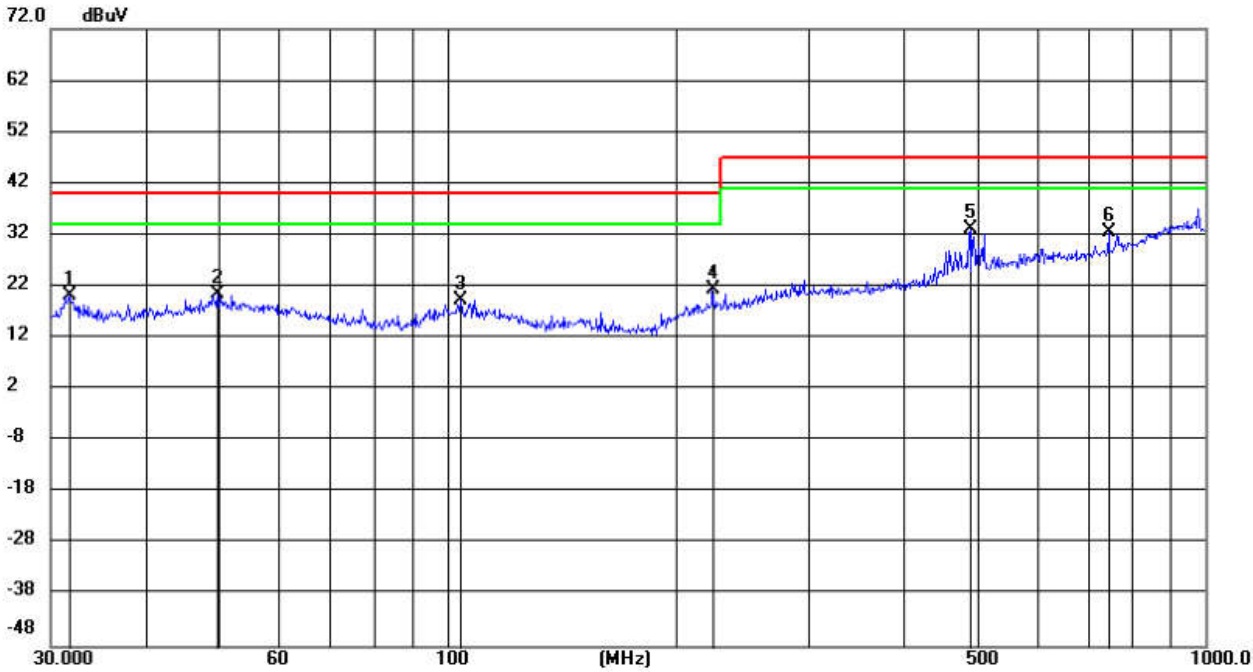
No.	Mk.	Freq.	Reading Level	Correct Factor	Measure-ment	Limit	Margin		Antenna Height	Table Degree	
		MHz	dBuV	dB	dBuV/m	dBuV/m	dB	Detector	cm	degree	Comment
1	*	53.5052	14.06	14.02	28.08	40.00	-11.92	QP	100	149	
2		99.8777	9.39	14.03	23.42	40.00	-16.58	QP	100	4	
3		232.5318	10.00	14.91	24.91	47.00	-22.09	QP	100	4	
4		365.5390	12.74	18.65	31.39	47.00	-15.61	QP	100	32	
5		560.6928	8.41	23.05	31.46	47.00	-15.54	QP	100	96	
6		744.8660	8.80	25.48	34.28	47.00	-12.72	QP	100	85	

Measurement Data（XIAO-ESP32-S3 ）：
Below 1GHz（QP）：
Horizontal



No.	Mk.	Freq.	Reading Level	Correct Factor	Measure-ment	Limit	Margin		Antenna Height	Table Degree	
		MHz	dBuV	dB	dBuV/m	dBuV/m	dB	Detector	cm	degree	Comment
1		49.7066	4.59	14.73	19.32	40.00	-20.68	QP	100	322	
2		108.6470	5.29	13.54	18.83	40.00	-21.17	QP	200	356	
3		159.7844	8.95	10.54	19.49	40.00	-20.51	QP	100	4	
4		239.9873	7.64	14.47	22.11	47.00	-24.89	QP	100	343	
5	*	400.4318	12.71	18.30	31.01	47.00	-15.99	QP	200	356	
6		470.5232	10.14	20.02	30.16	47.00	-16.84	QP	100	97	

Vertical



No.	Mk.	Freq.	Reading Level	Correct Factor	Measure-ment	Limit	Margin		Antenna Height	Table Degree	
		MHz	dBuV	dB	dBuV/m	dBuV/m	dB	Detector	cm	degree	Comment
1		31.7313	7.35	12.76	20.11	40.00	-19.89	QP	200	351	
2		49.7068	5.68	14.73	20.41	40.00	-19.59	QP	100	356	
3		104.1701	5.79	13.60	19.39	40.00	-20.61	QP	100	356	
4		223.7334	7.75	13.82	21.57	40.00	-18.43	QP	200	85	
5	*	489.0269	12.77	20.47	33.24	47.00	-13.76	QP	200	203	
6		744.8661	8.32	24.10	32.42	47.00	-14.58	QP	200	160	

Note:
1. Margin=Measurement-Limit.
2. Measurement=Reading_Level+Correct Factor.
3. Correct Factor=Ant Factor+Cable loss.
4. The disturbance above 1GHz was very low, more than 20dB below the limit, so only the below 1GHz had been displayed.

7.2 EMS (Immunity)

Performance Criteria of EN 301 489-17, sub clause 6.2 table 2.

Table 2: Performance criteria

Criteria	During test	After test (i.e. as a result of the application of the test)
A	Shall operate as intended. (See note). Shall be no loss of function. Shall be no unintentional transmissions.	Shall operate as intended. Shall be no degradation of performance. Shall be no loss of function. Shall be no loss of critical stored data.
B	May be loss of function.	Functions shall be self-recoverable. Shall operate as intended after recovering. Shall be no loss of critical stored data.
C	May be loss of function.	Functions shall be recoverable by the operator. Shall operate as intended after recovering. Shall be no loss of critical stored data.
NOTE: Operate as intended during the test allows a level of degradation in accordance with clause 6.2.2.		

7.2.1 Radiated Immunity

Test Requirement: EN 301 489-17 Clause 7.2
Test Method: EN 301 489-1 Clause 9.2.2

EUT Operation:
Ambient: Temp.: 22°C Humid.: 50% Press.: 1010 mbar
Test Mode: BT mode,WIFI mode and Standby mode
Criterion Required: A
Equipment Used: Refer to section 6 for details.
Test Setup:

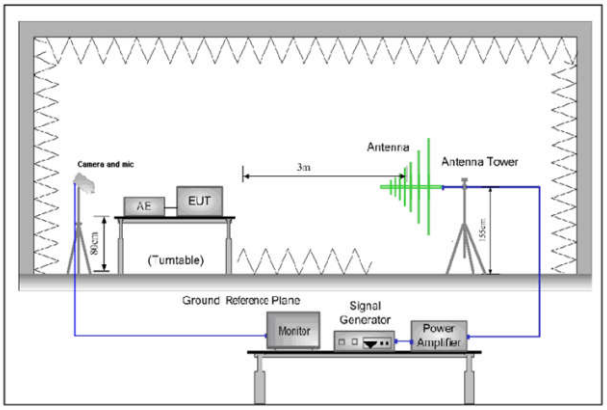


Figure 1. 80MHz to 1GHz

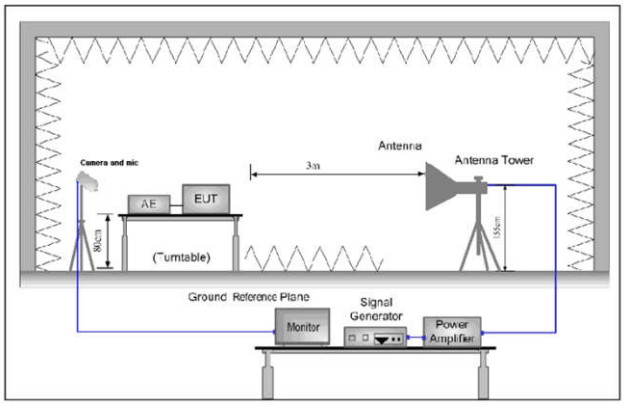


Figure 2. 1GHz to 6GHz

Test Procedure:

1) For table-top equipment, the EUT was placed in the chamber on a non-conductive table 0.8m high. For arrangement of floor-standing equipment, the EUT was mounted on a non-conductive support 0.1m above the supporting plane. For human body-mounted equipment, the EUT may be tested in the same manner as table top items.
2) If possible, a minimum of 1 m of cable is exposed to the electromagnetic field. Excess length of cables interconnecting units of the EUT shall be bundled low-inductively in the approximate center of the cable to form a bundle 30 cm to 40 cm in length.
3) The EUT was initially placed with one face coincident with the calibration plane. The EUT face being illuminated was contained within the UFA (Uniform Field Area).
4) The frequency ranges to be considered were swept with the signal modulated and pausing to adjust the RF signal level or to switch oscillators and antennas as necessary. Where the frequency range was swept incrementally, the step size was not exceed 1% of the preceding frequency value.
5) The dwell time of the amplitude modulated carrier at each frequency was not be less than the time necessary for the EUT to be exercised and to respond, and was not less than 0.5 s.
6) The test normally was performed with the generating antenna facing each side of the EUT.
7) The polarization of the field generated by each antenna necessitates testing each selected side twice, once with the antenna positioned vertically and again with the antenna positioned horizontally.
8) The EUT was performed in a configuration to actual installation conditions, a video camera and/or an audio monitor were used to monitor the performance of the EUT.

Test result: PASS

Test Data 1 （XIAO-ESP32-S3-Sense ）

Frequency	Level	Modulation	EUT Face	Antenna Polaxis	Result / Observations
80MHz-1GHz, 1GHz to 6GHz	3V/m	1kHz, 80% Amp. Mod, 1% increment Dwell time: 1 seconds	Front	V	A
				H	A
			Back	V	A
				H	A
			Left	V	A
				H	A
			Right	V	A
				H	A

Test Data 2 （XIAO-ESP32-S3）

Frequency	Level	Modulation	EUT Face	Antenna Polaxis	Result / Observations
80MHz-1GHz, 1GHz to 6GHz	3V/m	1kHz, 80% Amp. Mod, 1% increment Dwell time: 1 seconds	Front	V	A
				H	A
			Back	V	A
				H	A
			Left	V	A
				H	A
			Right	V	A
				H	A

Remark: A: No performance degradation during test.

7.2.2 ESD

Test Requirement: EN 301 489-17 Clause 7.2

Test Method: EN 301 489-1 Clause 9.3.2

EUT Operation:

Ambient: Temp.: 23°C Humid.: 54% Press.: 1010 mbar

Test Mode: BT mode and Standby mode

Criterion Required: B

Discharge Impedance: 330 Ω / 150 pF

Polarity: Positive & Negative

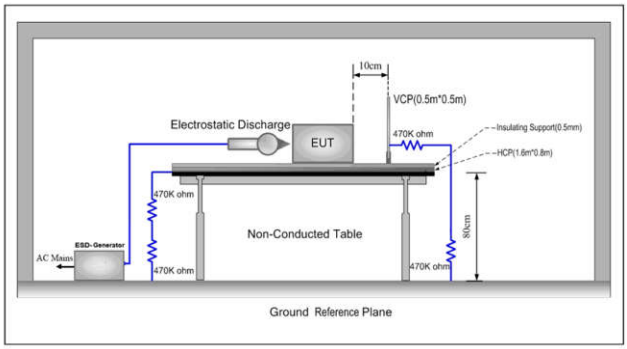
Number of Discharge: Minimum 10 times at each test point

Discharge Mode: Single Discharge

Discharge Period: 1 second minimum

Equipment Used: Refer to section 6 for details.

Test Setup:



Test set-up for tabletop equipment

Test Procedure:

1) Contact discharges to the conductive surfaces and to coupling planes:

The EUT was exposed to at least 200 discharges, 100 each at negative and positive polarity, at a minimum of four test points (a minimum of 50 discharges at each point). One of the test points was subjected to at least 50 indirect discharges (contact) to the centre of the front edge of the horizontal coupling plane. The remaining three test points shall each receive at least 50 direct contact discharges. If no direct contact test points were available, then at least 200 indirect discharges were applied in the indirect mode. Tests were performed at a maximum repetition rate of one discharge per second.

Air discharge at slots and apertures, and insulating surfaces:

On those parts of the EUT where it was not possible to perform contact discharge testing, the equipment was investigated to identify user accessible points where breakdown may occur. This investigation was restricted to those areas normally handled by the user. A minimum of 10 single air discharges were applied to the selected test point for each such area.

The application of electrostatic discharges to the contacts of open connectors was not required by this standard.

2) The EUT was put on a 0.8m high wooden table for table-top equipment or 0.1m high for floor standing equipment standing on the ground reference plane(GRP).

3) A horizontal coupling plane(HCP) 1.6m by 0.8m in size was placed on the table, and the EUT with its cables were isolated from the HCP by an insulating support thick than 0.5mm. The VCP 0.5m by 0.5m in size & HCP were constructed from the same material type & thinkmess as that of the GRP, and connected to the GRP via a 470kΩ resistor at each end. The distance between EUT and any of the other metallic surface excepted the GRP, HCP and VCP was greater than 1m.

4) During the contact discharges, the tip of the discharge electrode was touch the EUT before the discharge switch is operated. During the air discharges, the round

discharge tip of the discharge electrode was approached as fast as possible to touch the EUT.

5) After each discharge, the ESD generator was removed from the EUT, the generator was then retriggered for a new single discharge. For ungrounded product, a discharge cable with two resistances was used after each discharge to remove remnant electrostatic voltage. 10 times of each polarity single discharge were applied to HCP and VCP.

Test Results: PASS

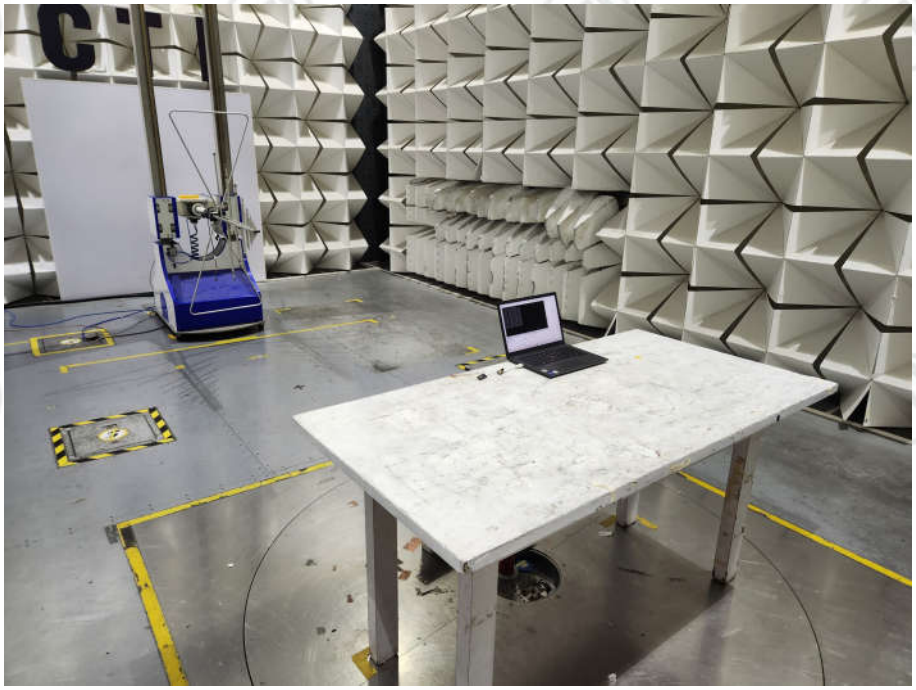
Observations:	Test Point: 1. All insulated enclosure and seams. 2. All accessible metal parts of the enclosure.			
Direct Application Test Results				
Direct Application			Test Results	
Discharge Level (kV)	Pulse No.	Test Point	Contact Discharge	Air Discharge
±8	10 for every level	1	N/A	N/A
±4	10 for every level	2	N/A	N/A
Indirect Application for tabletop equipment Test Results				
Indirect Application			Test Results	
Discharge Level (kV)	Pulse No.		Horizontal Coupling	Vertical Coupling
±4	10 for every level		A	A
Criterion Required:	B			

Remark:

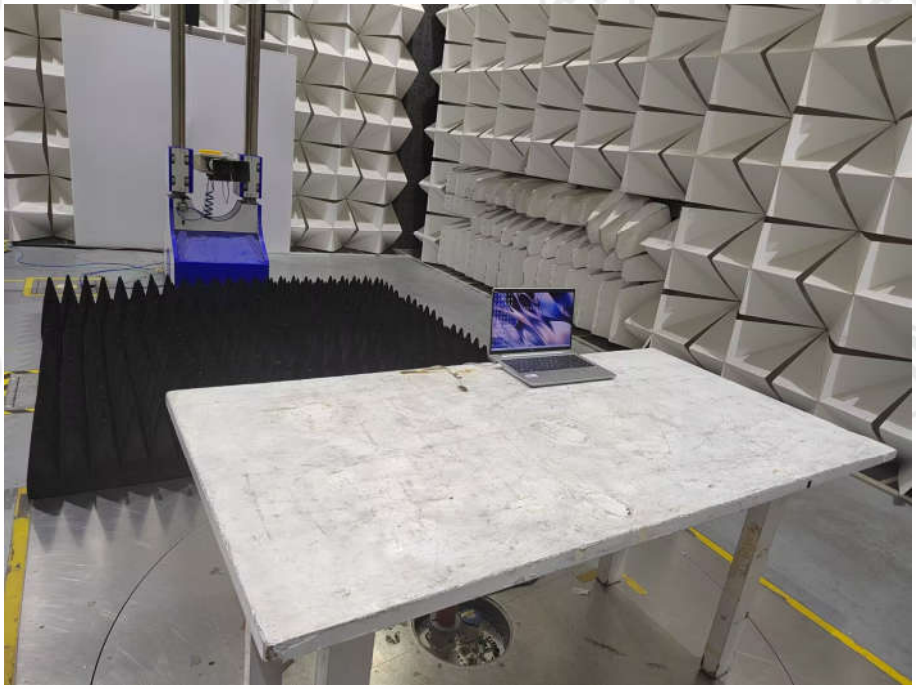
A: No performance degradation during test.

N/A: Not applicable.

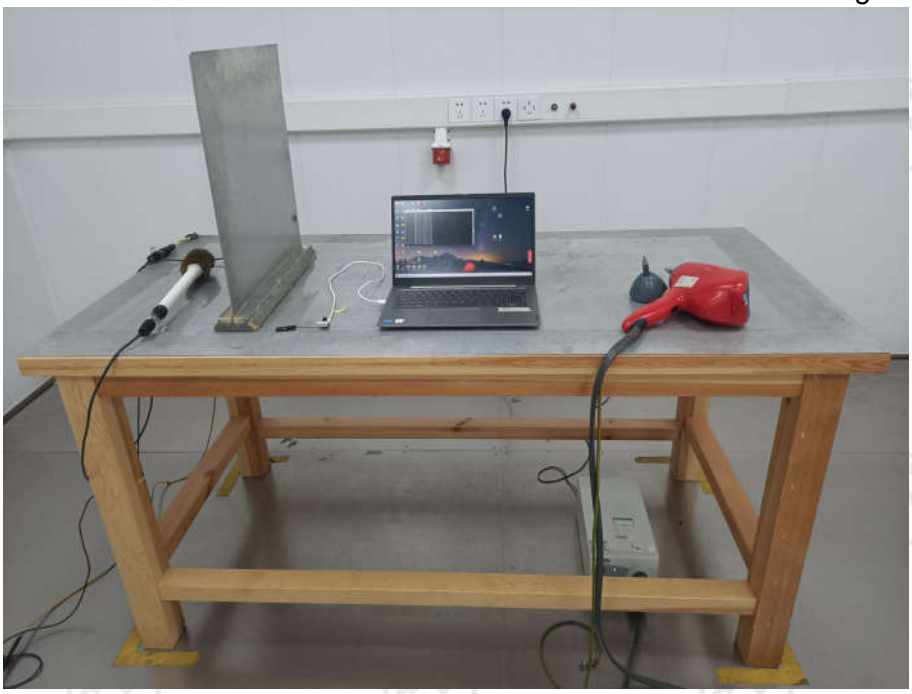
APPENDIX PHOTOGRAPHS OF TEST SETUP



RADIATED DISTURBANCE TEST SETUP-1

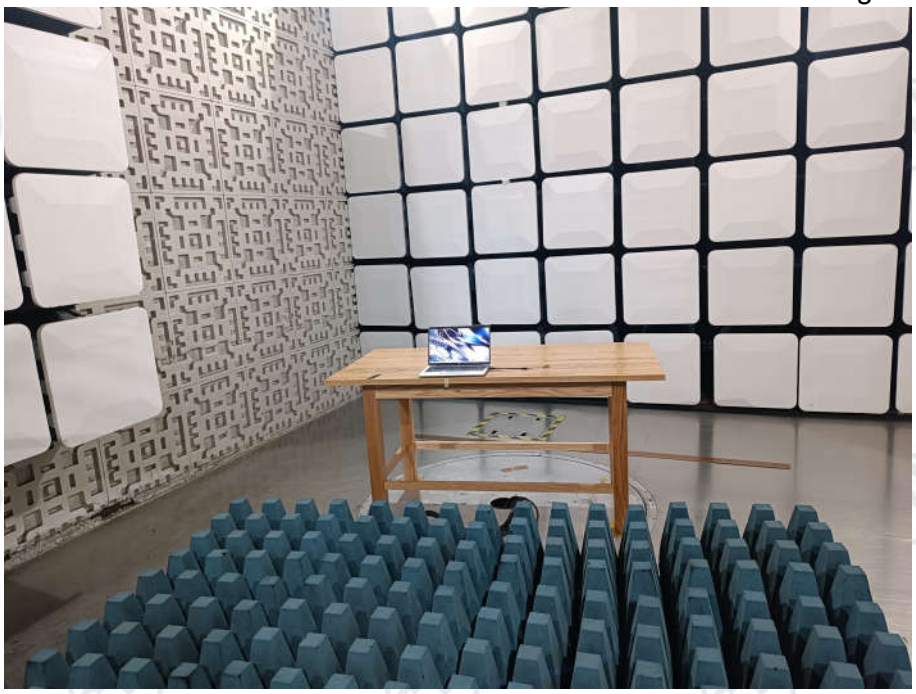


RADIATED DISTURBANCE TEST SETUP-2



ESD TEST SETUP-1

ESD TEST SETUP-2



RADIO-FREQUENCY ELECTROMAGNETIC FIELD TEST SETUP

APPENDIX PHOTOGRAPHS OF EUT

Refer to Report No. EED32P80440401 for EUT external and internal photos.

The test report is effective only with both signature and specialized stamp. The result(s) shown in this report refer only to the sample(s) tested. Without written approval of CTI, this report can't be reproduced except in full.

*** End of Report ***